Subtraction and Negative Signs: Diagnosing the Error

Before you compute, rewrite every subtraction as an addition:

- $p - q$ means $p + (-q)$ — "add the opposite" of the second number.
- Subtracting a negative adds a positive: $p - (-q) = p + q$, so the result moves *right* on the number line.
- Subtracting a positive adds a negative, so the result moves *left*.
- The sign of the answer follows the number with the larger distance from zero, not the one written first.

For every find-and-fix item: name the step where the sign was lost, then redo it correctly.

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1. Rewrite as an addition, then evaluate.

$$-8 - (-3)$$

Answer: ________________

2. Rewrite as an addition, then evaluate. Say in one phrase why the answer is negative even though the first number is positive.

$$5 - 12$$

Answer: ________________

3. Evaluate. Keep the decimals exact — do not round.

$$-3.5 - 2.25$$

Answer: ________________

4. Write the correct symbol (<, >, or =) in the blank, then finish the sentence below it.

$$-8 - (-3) \quad \underline{\hspace{2em}} \quad -8 + 3$$

Subtracting -3 has the same effect as ____________________ because ________________________.

Answer: ________________

5. Order these three values from **least to greatest**. Write the three results, not the expressions.

$$-2 - 7 \qquad 2 - 7 \qquad -2 + 7$$

Answer: ________________

6. Find and fix the mistake. Dana turns in this work:

$$-4 - (-9) = -13$$

Dana's answer is -13. Name the rule she broke, then write the correct value.

Answer: ________________

7. Find and fix the mistake. Marcus turns in this work:

$$-7 - 4 = -3$$

Marcus's answer is -3. Name the rule he broke, then write the correct value.

Answer: ________________

8. Challenge — find and fix. A student evaluates a three-term expression left to right and gets -12.75:

$$-2.5 - (-6) - 4.25$$

Identify the single sign the student mishandled, rewrite all of the subtractions as additions, and compute the correct value.

Answer: ________________